

A PROJECT REPORT ON
OPTIMIZING TRANSPORTATION FOR
SUSTAINABILITY

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FOR THE AWARD OF THE DEGREE

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CERTIFICATE

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ABSTRACT

The project Optimizing Transportation for Sustainability is a multi-facet project that provides suggestions, and strategies for optimizing transportation with an emphasis on sustainability, focusing on ride-sharing, ride-pooling, and the determination of the shortest routes. By integrating advanced data analytics, data structure algorithms, and innovative urban planning techniques, the project caters to minimizing the environmental impact of transportation systems while enhancing efficiency and accessibility. Key areas of the project include optimizing ride-sharing routes using A-star algorithm, with Manhattan-distance for providing accurate heuristics, encouraging carpooling initiatives by determining the riders that intersect the other route by custom developing algorithms to identify these in most efficient travel paths. The project employs a multi-disciplinary approach, drawing insights from environmental science, computer science, and urban planning to propose actionable recommendations. Preliminary results indicate significant potential for reducing carbon emissions, lowering transportation costs, and easing traffic congestion. This research underscores the critical aspects of achieving sustainable development goals and offers a framework for policymakers and urban planners to design more resilient and eco-friendly transportation networks.

Keywords:

Transportation Optimization, sustainability, Smart Cities, ride-sharing, graph theory, ride-pooling, shortest path algorithms, A-star algorithm, data analytics, urban planning, environmental impact, carbon emissions reduction.

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LIST OF ABBREVIATIONS

Abbreviation	Illustration
SaaS	Software as a Service
CO ₂	Carbon dioxide
GPS	Global Positioning System
KPI	Key Performance Indicator
UI	User Interface

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CHAPTER 1
INTRODUCTION

1.1 OVERVIEW

Transportation is a significant contributor to greenhouse gas emissions, making it crucial to adopt sustainable practices that reduce our carbon footprint. Sustainable transportation includes the use of alternative fuel sources, such as electric vehicles and biofuels, which produce fewer emissions compared to traditional fossil fuels. Enhancing public transportation systems, like buses and trains, can also decrease the number of single-occupancy vehicles on the road, leading to lower emissions and less traffic congestion.

Adding active modes of transport such as cycling and walking is another vital aspect of sustainable transport. By developing pedestrian-friendly infrastructure and dedicated cycling lanes, cities can encourage these eco-friendly options, promoting both environmental benefits and public health. Moreover, integrating smart technologies can optimize traffic management, reducing travel times and emissions.

Ultimately, embracing sustainable transportation practices is vital for fostering a healthier planet for future generations. By reducing emissions and promoting efficient transportation systems, we can combat climate change and enhance the quality of life in our communities.

1.2 MOTIVATION

Traffic congestion, energy shortages, and environmental pollution have become critical challenges for many metropolitan areas, significantly hindering their growth and quality of life. As urban populations rise, existing transportation infrastructures struggle to meet increasing demands, leading to severe traffic jams and lost productivity. This inefficiency not only frustrates commuters but also contributes to heightened stress levels among residents. Moreover, energy shortages exacerbate these issues, as cities rely heavily on fossil fuels, which are finite and subject to price volatility. Transitioning to sustainable transportation is essential to alleviate energy concerns by promoting renewable energy sources and reducing dependence on non-renewable fuels. Environmental pollution from vehicle emissions poses serious health risks, contributing to respiratory diseases and climate change. The urgent need for sustainable urban transportation has garnered attention from policymakers and citizens alike. By prioritizing sustainable practices, cities can improve air quality, reduce congestion, and foster a healthier environment, paving the way for long-term sustainability and resilience in urban development.

1.3 PROBLEM DEFINITION AND OBJECTIVES

Making transportation sustainable in smart cities is one of the most critical challenges of our time. Many issues related to transportation sustainability often overlook key factors contributing to CO₂ emissions. For instance, some approaches focus solely on vehicle operation attributes like running and idling, neglecting other significant elements such as vehicle type, age, and fitness. This narrow focus leads to an incomplete understanding of the sustainability challenges that urban transportation systems face.

To address these challenges, it is essential to holistically evaluate all factors that impact transportation sustainability, recognizing that a comprehensive approach is necessary for effective solutions. Additionally, it is crucial to examine how different vehicle types can contribute to further reductions in emissions. Without proper control of CO₂ emissions, deteriorating air quality can have severe health implications, complicating travel for urban residents. Therefore, addressing commuting challenges requires a broader perspective that goes beyond the mere availability of transportation. Solutions must also tackle issues such as closed routes, the optimization of travel paths, and the avoidance of congestion to create a sustainable and efficient urban transportation ecosystem.

1.4 PROJECT SCOPE & LIMITATIONS

The project **"Optimizing Transportation for Sustainability"** aims to develop a web-based application that optimizes transportation routes by integrating sustainability metrics. The primary focus areas include optimizing ride-sharing and ride-pooling routes, calculating the shortest paths in a transportation network, and providing actionable insights for urban planners and policymakers.

LIMITATIONS

- **Dependency on External APIs:** The effectiveness of the application relies on the availability and accuracy of data from third-party APIs, which may not always be reliable.
- **Internet Connectivity Required:** Users must have a stable internet connection to access the application, limiting usage in areas with poor connectivity.
- **Data Privacy Concerns:** Storing user location and personal data can raise privacy issues that need to be addressed through robust security measures.
- **Complexity of Algorithm:** The A* algorithm, while effective, may require significant computational resources, especially when handling large datasets or numerous simultaneous users.
- **Limited Scope of Sustainability Metrics:** While the system focuses on emissions and fuel consumption, it may not encompass all aspects of sustainability, such as social or economic factors.

1.5 METHODOLOGIES OF PROBLEM SOLVING

The issue of how to optimize transportation for sustainability is tackled through intelligent routing algorithms paired with real-time geographic information. The system also mainly relies on the Google Directions API, with its ability providing a solid core for calculating valid and efficient paths.

Directions API, internally, employs sophisticated pathfinding algorithms, specifically the A* (A-Star) search algorithm, to compute the shortest or most optimal route between two locations. A* unites the strengths of Dijkstra's Algorithm (optimality) and Greedy Best-First Search (efficiency) and is best suited for moving through dynamic urban road networks where the conditions are constantly changing.

In addition to shortest-path computations, the API includes real-time updates of road closures, incidents, construction, and traffic congestion. This allows the system to provide reliable alternate routes, reducing both delay and unnecessary vehicle emissions from rerouting or idling.

Forward-looking, the project seeks to improve this approach by incorporating sustainability-oriented logic into routing choice. Instead of choosing routes by distance or time alone, future enhancements will consider real-time CO emissions, fuel usage, and road quality when suggesting greener routes.

This vision can be achieved by:

- Maritime third-party emission and environmental data sources integration.
- Enhancing routing score calculation to include sustainability metrics.
- Providing user preference options for “eco-friendly routing” modes.

With these improvements, the system can go beyond efficiency to enable environmentally friendly transport decisions in smart cities.

CHAPTER 2

LITERATURE SURVEY

Title	Authors	Source	Shortcoming	Solution
A Real-Time Decision Support Approach for Managing Disruptions in Line-Haul Freight Transport Networks	Ahmed Karam and Kristian Hegner Reinau	IEEE Transactions on Intelligent Transportation Systems, Vol. 23, No. 12, December 2022	Focuses primarily on urban freight distribution; limited consideration of CO2 emissions and indirect costs of delays; reliance on human planners for decision-making	Proposes a hybrid approach using Discrete Event Simulation (DES), optimization algorithms, and cost-effectiveness analysis to manage disruptions effectively in line-haul transport networks.
A Decade Review on Smart Cities: Paradigms, Challenges and Opportunities	Tarana Singh, Arun Solanki, Sanjay Kumar Sharma, Anand Nayyar, Anand Paul	IEEE Access (2024)	Insufficient focus on sustainability and environmental impacts in smart city planning.	Emphasize sustainable practices in development, leveraging green technologies and eco-friendly solutions.
Cognitive City Platform as Digital Public Infrastructure for Developing a Smart, Sustainable and Resilient City in Indonesia	Okyza Maherdy Prabowo, Eueung Mulyana, I Gusti Bagus Baskara Nugraha, Suhono Harso Supangkat	IEEE Access (2023)	Limited adaptability of existing smart city platforms	A novel cognitive city platform architecture for diverse urban scales.
A Systematic Review of Analytic Hierarchy Process Applications to Solve Transportation Problems: From 2003 to 2022	Sarbast Moslem, Mahyar Kamali Saraji, Abbas Mardani, Ahmad Alkharabsheh, Szabolcs Duleba, Domokos Esztergár-Kiss	IEEE Access, Volume 11, 2023	Limited investigation into various aspects of transportation problems. Many AHP extensions have been developed, but some have not been applied in transportation. Lack of comprehensive review of AHP applications in transportation.	Analyzing the status quo of AHP applications to identify which extensions have not been applied. Future studies should explore more transportation problems that could be solved using AHP.

CHAPTER 3

SOFTWARE REQUIREMENTS SPECIFICATION

3.1 ASSUMPTIONS AND DEPENDENCIES

- Users will have reliable internet access to use the application.
- Users will have compatible devices to access the web application.
- The system relies on third-party APIs for real-time traffic and emissions data.

3.2 FUNCTIONAL REQUIREMENTS

3.2.1 SYSTEM FEATURE 1 (USER MANAGEMENT)

- **FR1:** The system shall allow users to create accounts.
- **FR2:** The system shall enable users to log in and log out.
- **FR3:** The system shall allow users to update their profile information.

3.2.2 SYSTEM FEATURE 2 (ROUTE OPTIMIZATION)

- **FR4:** The system shall provide an interface for users to input their start and end locations.
- **FR5:** The system shall calculate the shortest path using the A* algorithm.
- **FR6:** The system shall consider sustainability factors (e.g., emissions, fuel consumption) in route calculations.

3.2.3 SYSTEM FEATURE 3 (SUSTAINABILITY METRICS)

- **FR7:** The system shall display emissions data for each route.
- **FR8:** The system shall allow users to compare multiple routes based on sustainability metrics.

3.2.4 SYSTEM FEATURE 4 (DATA VISUALIZATION)

- **FR9:** The system shall provide interactive maps to visualize routes.
- **FR10:** The system shall generate reports on emissions and travel times.

3.2.5 SYSTEM FEATURE 5 (CARBON FOOTPRINT ESTIMATION)

- **FR11:** The system includes a carbon footprint estimator based on the user's lifestyle and travel habits.
- **FR12:** Users input data such as daily transport usage, electricity consumption, flight frequency, and diet type.
- **FR13:** The system calculates monthly CO emissions and presents results using graphs and charts.

3.2.6 SYSTEM FEATURE 6 (REAL-TIME ECO-ROUTE RECOMMENDATION)

- **FR14:** The system calculate and show real-time route options (eco-friendly, fastest, balanced) using the A * algorithm with a modified cost function.
- **FR15:** Users input source, destination, vehicle and fuel type, and travel preference.
- **FR16:** The system returns route options with distance, time, emissions, fuel consumption, and cost breakdown.

3.2.7 SYSTEM FEATURE 7 (ORGANIZATION-BASED MODULE)

- **FR17:** The system supports organization-specific dashboards with role-based login (Admin vs Employee).
- **FR18:** Organizations can register, manage users, assign challenges, and monitor emissions.
- **FR19:** Admin dashboards display department-wise emissions, vehicle usage trends, and allow CSV/PDF report downloads.

3.3 EXTERNAL INTERFACE REQUIREMENTS

3.3.1 USER INTERFACES

The application will feature a web-based user interface with a responsive design for both desktop and mobile devices.

3.3.2 HARDWARE INTERFACES

The system will interface with GPS devices to enable real-time location tracking for route optimization.

3.3.3 SOFTWARE INTERFACES

Integration with third-party APIs for real-time traffic updates and emissions data, including but not limited to Google Maps API and other transportation data services.

3.3.4 COMMUNICATION INTERFACES

The system will communicate over standard web protocols (HTTP/HTTPS) to interact with users and third-party services.

3.4 NONFUNCTIONAL REQUIREMENTS

3.4.1 PERFORMANCE REQUIREMENTS

- **NFR1:** The system shall calculate routes within 5 seconds.
- **NFR2:** The system shall handle up to 1,000 simultaneous users without performance degradation.

3.4.2 SAFETY REQUIREMENTS

The application shall ensure data integrity and user privacy, particularly concerning location and personal information.

3.4.3 SECURITY REQUIREMENTS

- **NFR3:** The system shall encrypt user data and passwords.
- **NFR4:** The system shall implement role-based access control for different user types.

3.4.4 SOFTWARE QUALITY ATTRIBUTES

- **SQA1:** Functional Correctness.
- **SQA2:** Performance and Scalability.
- **SQA3:** No Regression.

3.5 SYSTEM REQUIREMENTS

3.5.1 DATABASE REQUIREMENTS

A relational database (Apache Derby) is used to store user data, route information, and sustainability metrics. By using the sqlite3 library, the extended module accesses the SQLite database to store, retrieve, update, and delete data from the SQLite file.

3.5.2 SOFTWARE REQUIREMENTS (PLATFORM CHOICE)

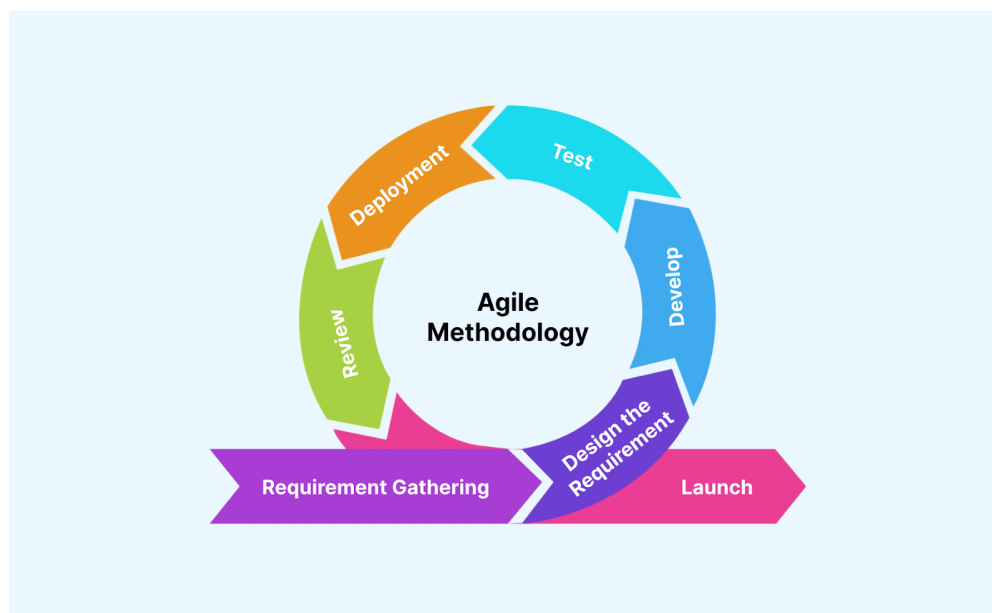
- **Operating System:** Windows Server or Linux.
- **Web Server:** Apache or Nginx, Flask's built-in development server to host the Python backend locally
- **Programming Languages:** Spring Boot, Java, HTML5, JavaScript, Python, Dart, Flutter.

3.5.3 HARDWARE REQUIREMENTS

A server with a minimum of 8 GB RAM, 100 GB disk space, and a multi-core processor to handle the application workload.

3.6 ANALYSIS MODEL: SDLC MODEL

The project will follow the **Agile Software Development Life Cycle (SDLC)** model to ensure flexibility, iterative development, and active stakeholder involvement throughout the project lifecycle.

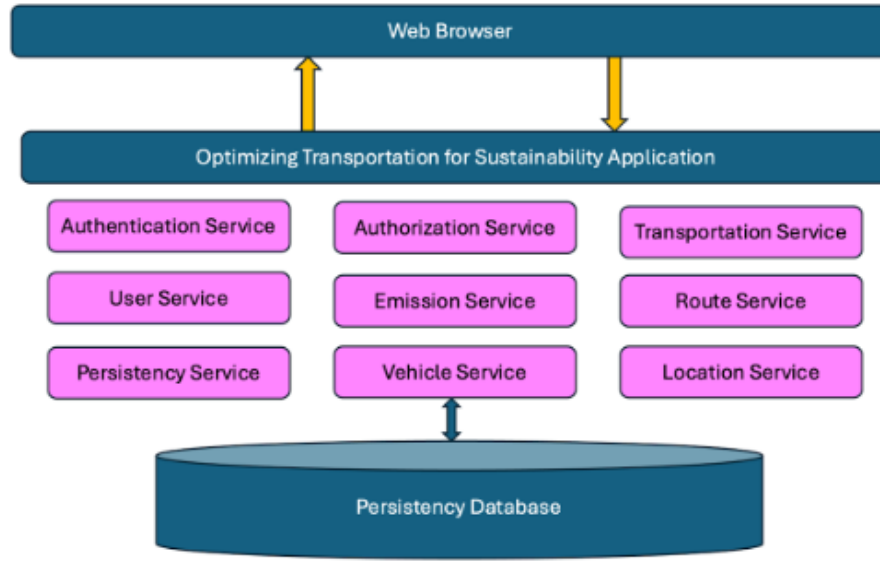


In Agile, the requirements are gathered by having continuous communication with the stakeholders such that the team knows what the users require and can incorporate the feedback at the earliest. The requirements are continuously reduced to user stories and prioritized in the product backlog. Lightweight and adaptable design patterns are applied during the design phase to prototype and test the main features quickly prior to proceeding to development. Development cycle is being divided into extremely small sprints, and gradually and incrementally features are implemented to speed up the delivery of working modules. Continuous integration and versioning are applied for code quality and consistency. It is tested in parallel with unit testing, integration tests, and acceptance testing by end-users at the completion of each sprint so that detection and fixing of issues can occur early. Once the sprint deliverables meet the acceptance tests, they are promoted to the staging or production environments for further consideration. Retrospectives and sprint reviews at fixed periods help in checking progress, incorporating learning experiences, and planning subsequent iterations better. The feedback loop allows the project to adapt to changing requirements, be of high quality, and ship a product that meets user expectations at the time of release.

CHAPTER 4

SYSTEM DESIGN

4.1 SYSTEM ARCHITECTURE



4.2 MATHEMATICAL MODEL

The cost function including emissions and fuel cost is:

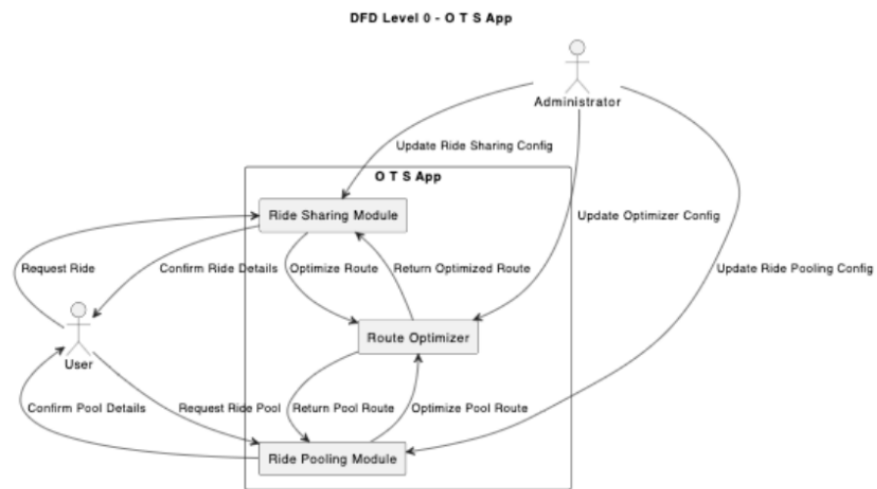
$$C_{ij} = \alpha \cdot T_{ij} + \beta \cdot E_{ij} + \gamma \cdot F_{ij}$$

where:

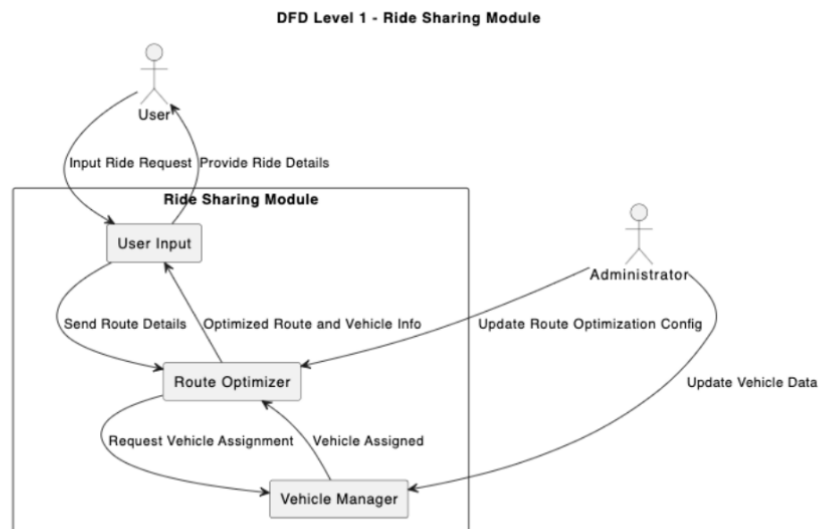
- T_{ij} is the distance cost [1 unit/km].
- E_{ij} is the emissions.
- F_{ij} is the fuel consumption.
- α , β , and γ are weighting factors for travel time, emissions, and fuel consumption, respectively. The sum of all factors should equal 1.

4.3 DATA FLOW DIAGRAM

4.3.1 DFD0 Diagram



4.3.2 DFD1 Diagram

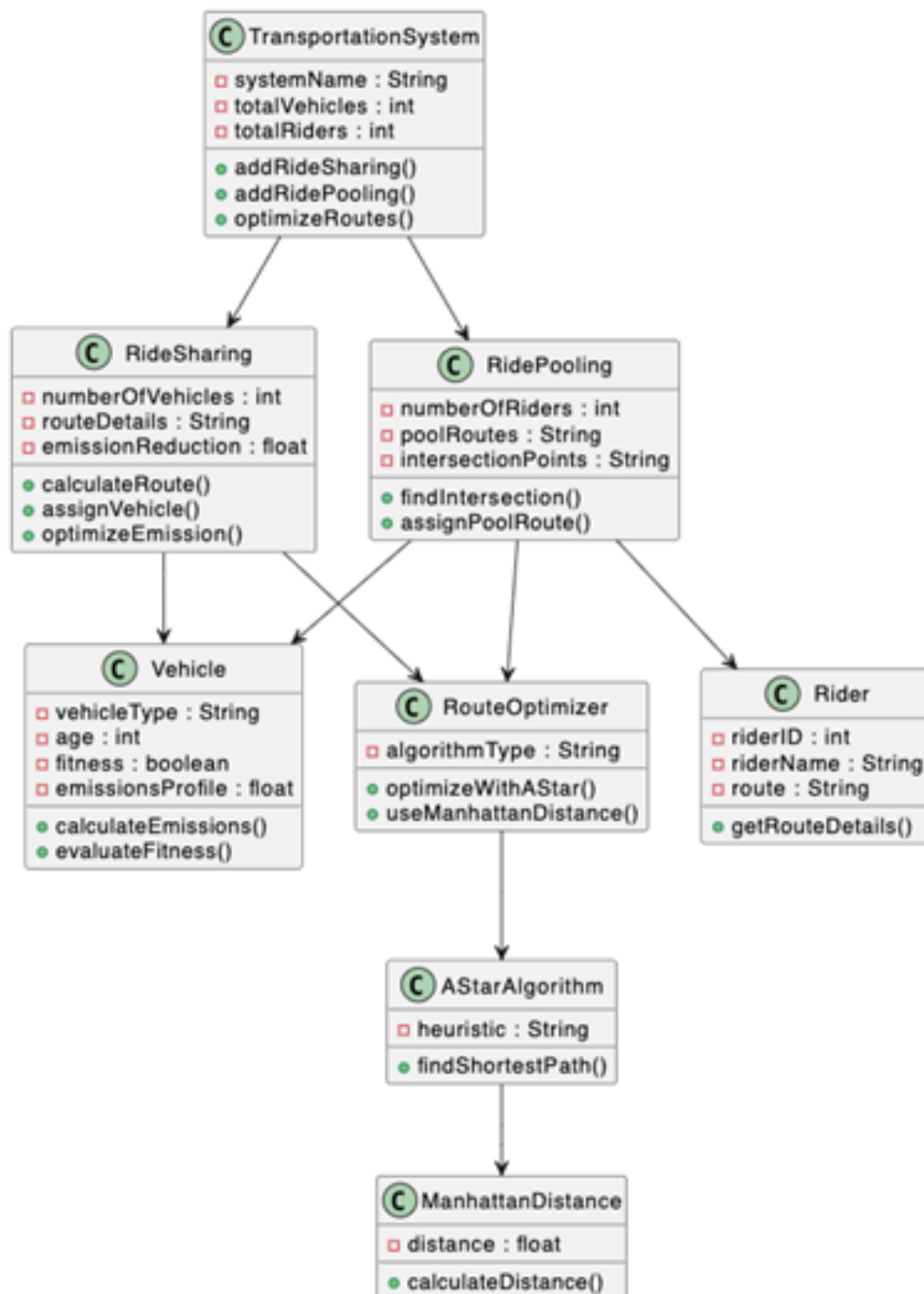


4.4 ENTITY RELATIONSHIP DIAGRAM

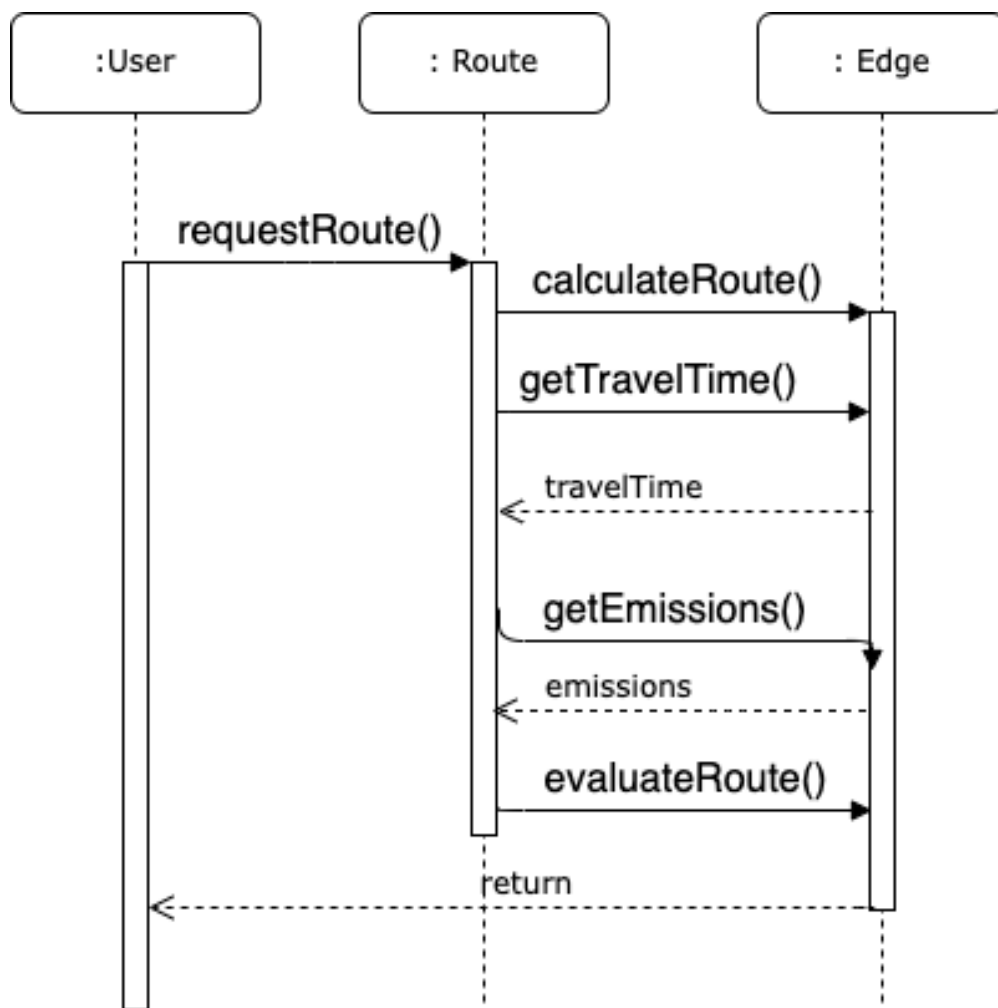


4.5 UML DIAGRAMS

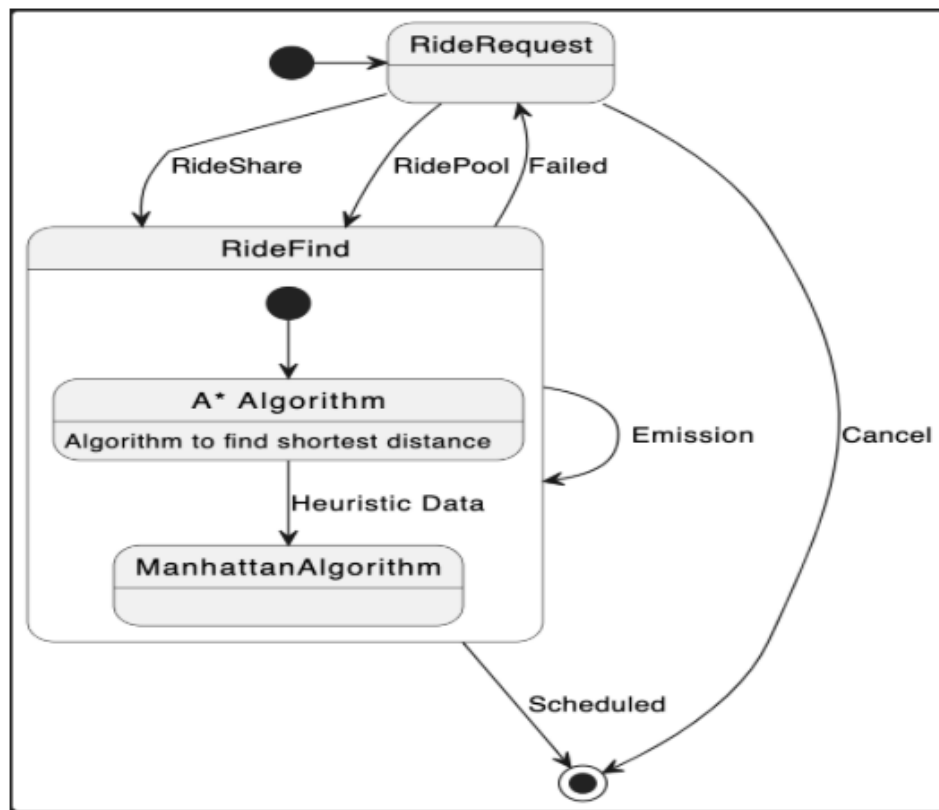
4.5.1 Class Diagram



4.5.2 Sequence Diagram



4.5.3 State Diagram



CHAPTER 5

PROJECT PLAN

5.1 PROJECT ESTIMATES

Phase	Duration
Initiation	1 Week
Planning	3 Weeks
Requirements	4 Weeks
Design	5 Weeks
Development	22 Weeks
Testing	12 Weeks
Deployment	3 Weeks
Maintenance	Ongoing

5.1.1 RECONCILED ESTIMATES

For our project, both development and deployment timelines can be estimated based on current resources and available technologies, with some inherent uncertainties.

5.1.1.1 Development Phase (Estimated Duration: 15-20 Weeks)

- **Requirement Analysis & Research (2-3 weeks):**
 - Understanding transportation sustainability goals.
 - Gathering data on transportation patterns, emissions, vehicle types, and traffic congestion.
 - Defining the scope, objectives, and key performance indicators (KPIs) for optimizing transportation.
- **System Architecture Design (2-3 weeks):**
 - Designing the overall system architecture, including cloud infrastructure, databases, and APIs.
 - Determining how smart technologies and optimization algorithms will be integrated.
 - Selecting tools, frameworks, and services provided by the chosen cloud platform (e.g., Azure AI services, AWS Lambda, or GCP AI solutions).
- **Data Collection & Preprocessing (3-4 weeks):**
 - Collecting real-time data from traffic sensors, vehicle GPS systems, and public transport feeds.
 - Cleaning and preprocessing the data for analysis (e.g., removing anomalies, handling missing values).
 - Integrating data sources into cloud storage (e.g., Azure Blob Storage, AWS S3, GCP Cloud Storage).

- **Algorithm Development & Testing (4-6 weeks):**
 - Developing optimization algorithms to reduce travel time, minimize emissions, and manage congestion.
 - Testing algorithms with historical data and simulated environments to ensure accuracy.
 - Iterating on algorithms based on testing results and performance metrics.
- **Frontend & Backend Development (4-5 weeks):**
 - Building a user-friendly interface for traffic managers and city planners to visualize insights.
 - Developing backend services to manage data flow, user access, and analytics.
 - Integrating APIs for real-time traffic data and vehicle monitoring.
- **Integration & Testing (2-3 weeks):**
 - Integrating all system components (algorithms, frontend, backend) and testing end-to-end functionality.
 - Conducting stress testing and performance benchmarking on cloud services to ensure scalability.

5.1.1.2 Deployment Phase (Estimated Duration: 2 Weeks)

- **Cloud Infrastructure Setup (1 week):**
 - Setting up the cloud infrastructure on Microsoft Azure, AWS, or GCP.
 - Configuring virtual machines, containers (e.g., Docker), or serverless environments (e.g., AWS Lambda, Azure Functions).
 - Setting up databases (e.g., Azure SQL Database, Amazon RDS, GCP BigQuery) and networking services.
- **Application Deployment & Configuration (3-5 days):**
 - Deploying the application onto the cloud environment.
 - Configuring continuous integration and deployment (CI/CD) pipelines for automated updates and monitoring.
 - Securing the application with firewalls, identity access management (IAM), and encryption services provided by the cloud provider.
- **Post-Deployment Testing & Monitoring (2-3 days):**
 - Running final checks to ensure all components are working as expected in the cloud environment.
 - Monitoring the application for performance, ensuring it scales properly and remains responsive during real-time use.
 - Implementing alerting mechanisms to notify teams of any issues (e.g., Azure Monitor, AWS CloudWatch, GCP Operations Suite).

By following this timeline, the project should be successfully developed in around **15-20 weeks** and deployed by the end of **2 weeks** using hyper scaler cloud platforms like **Azure, GCP, or AWS**. This allows flexibility in scaling, monitoring, and making future optimizations.

5.1.2 PROJECT RESOURCES

A 3-member team assuming the roles of **Developer (front-end, back-end), Quality Assurance**.

Software and hardware resources leveraged by personal computers and laboratory equipments.

5.2 RISK MANAGEMENT

Effective risk management is essential for ensuring the success of our project, "**Optimizing Transportation for Sustainability**." This involves identifying potential risks, analyzing their impact, and implementing strategies to mitigate, monitor, and manage these risks.

5.2.1 RISK IDENTIFICATION

Risk identification involves recognizing potential challenges that could impact the development, deployment, and operation of the project. Key risks for this project include:

- **Technical Risks:**
 - **License Availability:** Difficulty in obtaining licenses for essential tools like databases, Google Maps API, and other third-party services that are integral to the project.
 - **Cloud Platform Reliability:** Dependency on hyperscalers (**Microsoft Azure, AWS, GCP**) may lead to service outages, which can affect real-time transportation optimizations.
 - **Integration Risks:** Ensuring that various components, such as data collection from traffic sensors, algorithms, and user interfaces, work seamlessly together.
- **Operational and Business Risks:**
 - **None Identified:** Since the project leverages cloud hyperscalers and existing APIs, no significant operational or business risks have been identified at this stage.
- **Schedule Risks:**
 - **Timelines:** There is a risk that the full scope of the project might not be achieved within the available timelines. While the minimum viable product (MVP) can be delivered on time, completing all desired features and integrations may be delayed.

5.2.2 RISK ANALYSIS

In risk analysis, we assess the likelihood and potential impact of identified risks on the project.

- **Technical Risks:**

- **Likelihood:** Moderate to high, depending on the availability and stability of third-party APIs and cloud services.
- **Impact:** High, as the failure or unavailability of essential APIs (e.g., Google Maps) can halt functionality related to route optimization or traffic prediction. Security breaches could result in data leaks or compliance issues.

- **Operational and Business Risks:**

- **Likelihood:** Low, as the operational aspects of this project depend primarily on technical success rather than external business operations.
- **Impact:** Moderate. Regulatory changes in transportation policies may require reworking certain aspects of the system.

- **Schedule Risks:**

- **Likelihood:** Moderate, as some aspects of the project may take longer than anticipated due to technical roadblocks or delays in license acquisitions.
- **Impact:** High. If certain milestones are delayed, only a minimal viable product (MVP) may be delivered within the expected timeframe, leaving the full scope at risk.

5.2.3 OVERVIEW OF RISK MITIGATION, MONITORING, MANAGEMENT

- **Technical Risk Mitigation:**

- Use of **REST APIs** for modular development of backend services, allowing for flexibility and easier troubleshooting of specific modules.
- Engaging with cloud platforms like **AWS**, **Azure**, or **GCP**, which provide a wide array of APIs, databases, and third-party services, reducing dependency on individual services like Google Maps.
- Regular code reviews and penetration testing to ensure robust security measures and data privacy protocols.

- **Availability of Licenses:**

- Early procurement of licenses for crucial services such as the **Google Maps API** and any necessary databases (e.g., **Azure SQL**, **Amazon RDS**) to avoid delays in integration.
- Backup options such as **OpenStreetMap** for route and traffic management in case Google Maps API becomes unavailable.

- **Operational and Business Risk Mitigation:**

- No significant operational or business risks identified. Monitoring for changes in government regulations that may impact the transportation landscape will be ongoing.
- **Schedule Risk Mitigation:**
 - Prioritizing the completion of critical backend functionality through **REST APIs**, ensuring that key features (e.g., route optimization, emissions tracking) are operational within the MVP.
 - Weekly sprints and milestone tracking to monitor progress and quickly address any delays in development or integration. Adjusting project scope or reprioritizing tasks based on available timelines.

5.3 PROJECT SCHEDULE

5.3.1 PROJECT TASK SET

Tasks and Milestones

1. Initiation

Objective: Define project scope, goals, and stakeholders.

- **1.1 Identify Stakeholders:** Week 1
- **1.2 Develop Project Charter:** Week 1-2
- **1.3 Define Project Objectives and Deliverables:** Week 1-2
- **1.4 Conduct Initial Feasibility Study:** Week 1-2
- **Milestone:** Project Charter and Feasibility Report Completed

2. Planning

Objective: Develop a comprehensive project plan.

- **2.1 Define Project Timeline and Milestones:** Week 3-5
- **2.2 Allocate Resources and Assign Roles:** Week 3-5
- **2.3 Develop Risk Management Plan:** Week 3-5
- **2.4 Create Detailed Work Breakdown Structure (WBS):** Week 3-5
- **2.5 Define Data Requirements and Sources:** Week 3-5
- **Milestone:** Project Plan, WBS, and Risk Management Plan Completed

3. Requirements Analysis

Objective: Gather and analyze requirements.

- **3.1 Conduct Stakeholder Interviews and Surveys:** Week 6-9
- **3.2 Define Functional and Non-Functional Requirements:** Week 6-9
- **3.3 Create User Personas and Use Cases:** Week 6-9
- **Milestone:** Requirements Specification Document Completed

4. Design

Objective: Design system architecture, database schema, and UI.

- **4.1 Develop System Architecture Diagram:** Week 10-14
- **4.2 Design Database Schema:** Week 10-14
- **4.3 Create UI Wireframes and Mock-ups:** Week 10-14
- **Milestone:** Design Documents Completed

5. Development

Objective: Develop the web-based application.

- **5.1 Set Up Development Environment:** Week 15
- **5.2 Implement Ride-Sharing Route Optimization:** Week 16-22
- **5.3 Implement Carpooling Algorithms:** Week 23-28
- **5.4 Integrate Data Analytics and Urban Planning Insights:** Week 29-32
- **5.5 Conduct Unit and Integration Testing:** Week 33-36

- **Milestone:** Core Features Implemented and Tested

6. Testing

Objective: Ensure application meets requirements.

- **6.1 Conduct System and Acceptance Testing:** Week 37-41
- **6.2 Perform User Acceptance Testing (UAT):** Week 42-44
- **6.3 Fix Bugs and Optimize Performance:** Week 45-48
- **Milestone:** Testing Completed and Application Ready for Deployment

7. Deployment

Objective: Deploy the application to production.

- **7.1 Set Up Production Environment:** Week 49
- **7.2 Deploy the Application:** Week 50
- **7.3 Conduct Final Testing and Validation:** Week 51-52
- **Milestone:** Deployed application

8. Maintenance and Evaluation

Objective: Maintain the application and evaluate its impact.

- **8.1 Monitor Application Performance:** Ongoing
- **8.2 Collect User Feedback and Usage Data:** Ongoing
- **8.3 Implement Updates and Improvements:** Ongoing
- **8.4 Evaluate Environmental Impact and Efficiency Improvements:** Ongoing
- **Milestone:** Regular Updates and Impact Evaluation Reports

5.3.2 TASK NETWORK

- **Phase 1: Requirement Gathering & Analysis**
 - **Tasks:** Collect data, define project scope, and identify sustainability goals.
 - **Dependencies:** No direct dependencies; forms the foundation for the project.
 - **Milestone:** Requirements documentation completed.
- **Phase 2: System Architecture & Design**
 - **Tasks:** Design system architecture (cloud, database, APIs), select cloud platforms (AWS, GCP).
 - **Dependencies:** Requires completed requirement analysis.
 - **Milestone:** System architecture finalized.
- **Phase 3: Data Collection & Preprocessing**
 - **Tasks:** Gather and clean transportation data, store in cloud databases.
 - **Dependencies:** Dependent on system design.
 - **Milestone:** Data ready for algorithm development.
- **Phase 4: Algorithm Development**
 - **Tasks:** Develop route optimization and emissions tracking algorithms.
 - **Dependencies:** Requires processed data.
 - **Milestone:** Initial algorithm tests passed.
- **Phase 5: Backend & Frontend Development**
 - **Tasks:** Build REST APIs and user interface.
 - **Dependencies:** Based on system architecture.
 - **Milestone:** Backend and frontend services completed.
- **Phase 6: Integration and Testing**
 - **Tasks:** Integrate backend, algorithms, and frontend; perform testing.
 - **Dependencies:** Requires completed backend and algorithms.
 - **Milestone:** System ready for deployment after passing tests.
- **Phase 7: Deployment**
 - **Tasks:** Deploy on cloud (AWS, Azure), set up monitoring.
 - **Dependencies:** Post-integration completion.
 - **Milestone:** System live and monitored.
- **Phase 8: Monitoring & Maintenance**
 - **Tasks:** Monitor performance, maintain system, collect user feedback.
 - **Dependencies:** Post-deployment.
 - **Milestone:** Continuous updates and feedback loop.
- **Critical Path**
 - System Architecture → Algorithm Development → Integration → Deployment.

5.3.3 TIMELINE CHART

Phase	Duration	Timeframe (Weeks)	Milestone
Requirement Gathering & Analysis	3 weeks	Week 1 - Week 3	Project requirements documented
System Architecture & Design	3 weeks	Week 4 - Week 6	System architecture finalized
Data Collection & Preprocessing	4 weeks	Week 7 - Week 10	Data prepared for algorithm development
Algorithm Development	6 weeks	Week 11 - Week 16	Algorithms tested and verified
Backend & Frontend Development	5 weeks	Week 17 - Week 21	Backend and frontend completed
Integration and Testing	4 weeks	Week 22 - Week 25	System passes testing
Deployment	2 weeks	Week 26 - Week 27	System deployed and live
Monitoring & Maintenance	Ongoing	Week 28 onwards	Continuous monitoring and updates

5.4 TEAM ORGANIZATION

5.4.1 TEAM STRUCTURE

The team consists of one UI Developer, one Back-end Developer, one Full-Stack Developer who also acts as Project Supervisor/Coordinator/Scrum Master, and one Quality Engineer.

Name	Role	Responsibility
Rajlaxmi Shrirame	Front-end Developer	Design UI mock-ups in Figma, develop the UI, integrate it with frontend, and follow HTML5 guidelines.
Vishakha Mishra	Full-Stack Developer, Project Manager, Scrum Master	Develop UI and Backend, integrate both. Supervise, coordinate and manage the project as Scrum Master.
Sneha Ghone	Quality Assurance Tester	Design, develop, and test REST APIs, along with testing and quality assurance of the project.

5.4.2 MANAGEMENT REPORTING AND COMMUNICATION

A Scrum team was established to ensure effective inter-team communication and smooth project coordination. We conducted daily Scrum calls to review progress, discuss road-blocks, and collaboratively plan tasks. These discussions were focused and constructive, promoting transparency, accountability, and team synergy. We regularly communicated our progress to our internal guide, Prof. N.Y. Kapadnis, through structured reports that highlighted both our achievements and the challenges faced. Additionally, we underwent four internal reviews where our work was assessed by Prof. N.Y. Kapadnis and two external guides. This structured communication and review framework ensured alignment with project goals, facilitated timely issue resolution, and maintained consistent progress throughout the project.

CHAPTER 6

PROJECT IMPLEMENTATION

6.1 OVERVIEW OF PROJECT MODULES

The project is written using the Spring Boot framework in Java, with the following modules that follow the MVC design pattern:

- **Rider Module**
 - This module registers riders in the application. The rider could be an administrator, ride owner, or simply a ride seeker.
- **Vehicle Module**
 - This module registers all the vehicles used for rides. It maintains records of all types of vehicles like Cars, Buses, Scooters, etc., along with their corresponding CO2 emissions.
- **Ride Module**
 - The Ride module allows the ride owner to give a ride or the ride seeker to take a ride. It provides a list of sustainable ride options to be picked from.
- **Emission Dashboard Module**
 - The Dashboard module aggregates various metrics and shows the vehicles, riders, rides, and emissions data.
- **Carbon Footprint Calculator**
 - This module estimates weekly CO2 emissions from travel and suggests eco-friendly alternatives to reduce the carbon footprint.

6.2 TOOLS AND TECHNOLOGIES USED

The core application is based on Visual Studio Code (VS Code) extended with Spring Boot, Apache Derby database, JUnit, and Selenium. VS Code is utilized as a light development environment, while Spring Boot is used to develop web applications. Apache Derby is used as the embedded database, and JUnit for unit testing. Selenium is applied to automate front-end testing through mimicking user inputs. All the above tools assist in effective development, testing, and deployment.

The extended module is developed using Flutter, Flask, and SQLite. Front-end development is performed using Flutter, which supports high-performance graphics, rich plugin support, and a hot-reload feature that enhances the productivity of developers. Flask is used for back-end development and offers flexibility with full control over server-side logic without exposing developers to complex configurations. SQLite is the light, open-source relational database that offers seamless integration through libraries and provides good support for dealing with application data. All of these technologies altogether offer an effective, responsive, and manageable extension of the central application.

6.3 ALGORITHM DETAILS

The project employs the A* search algorithm with the Manhattan heuristic for determining the shortest path between two points. The Manhattan heuristic computes the distance as the sum of horizontal and vertical displacements. Combined with the Google Directions and Map APIs, the system accesses real-time route data and traffic information, making dynamic and correct path computation possible. Optimal routing under prevailing conditions is thus guaranteed.

CHAPTER 7

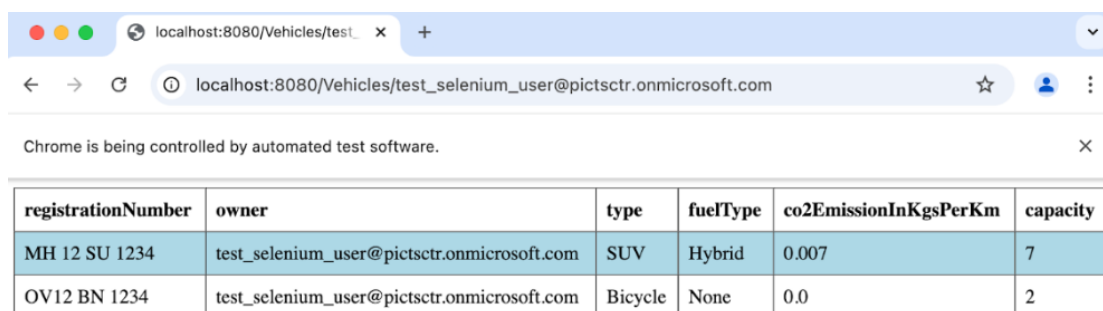
SOFTWARE TESTING

7.1 TYPE OF TESTING

- **Unit Testing :** Using the JUnit framework, each module is tested to ensure there are no compile-time or runtime errors. A manual round of testing is also performed, together with automation.
- **Integration Testing :** Selenium-based automation testing is done to ensure that each module interacts with each other seamlessly (e.g., a rider can give a ride only when the rider and vehicle are registered, etc.).
- **Performance Testing :** Performance under load is tested using both Selenium and manual testing to see how the system scales when subjected to stress, such as creating thousands of riders, etc.

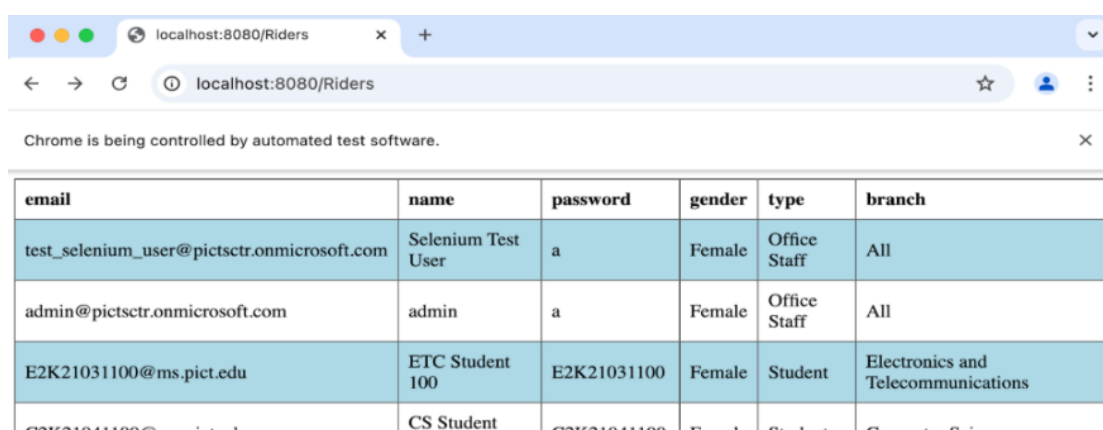
7.2 TEST CASES & TEST RESULTS

- **View Vehicles Selenium Automated Test**



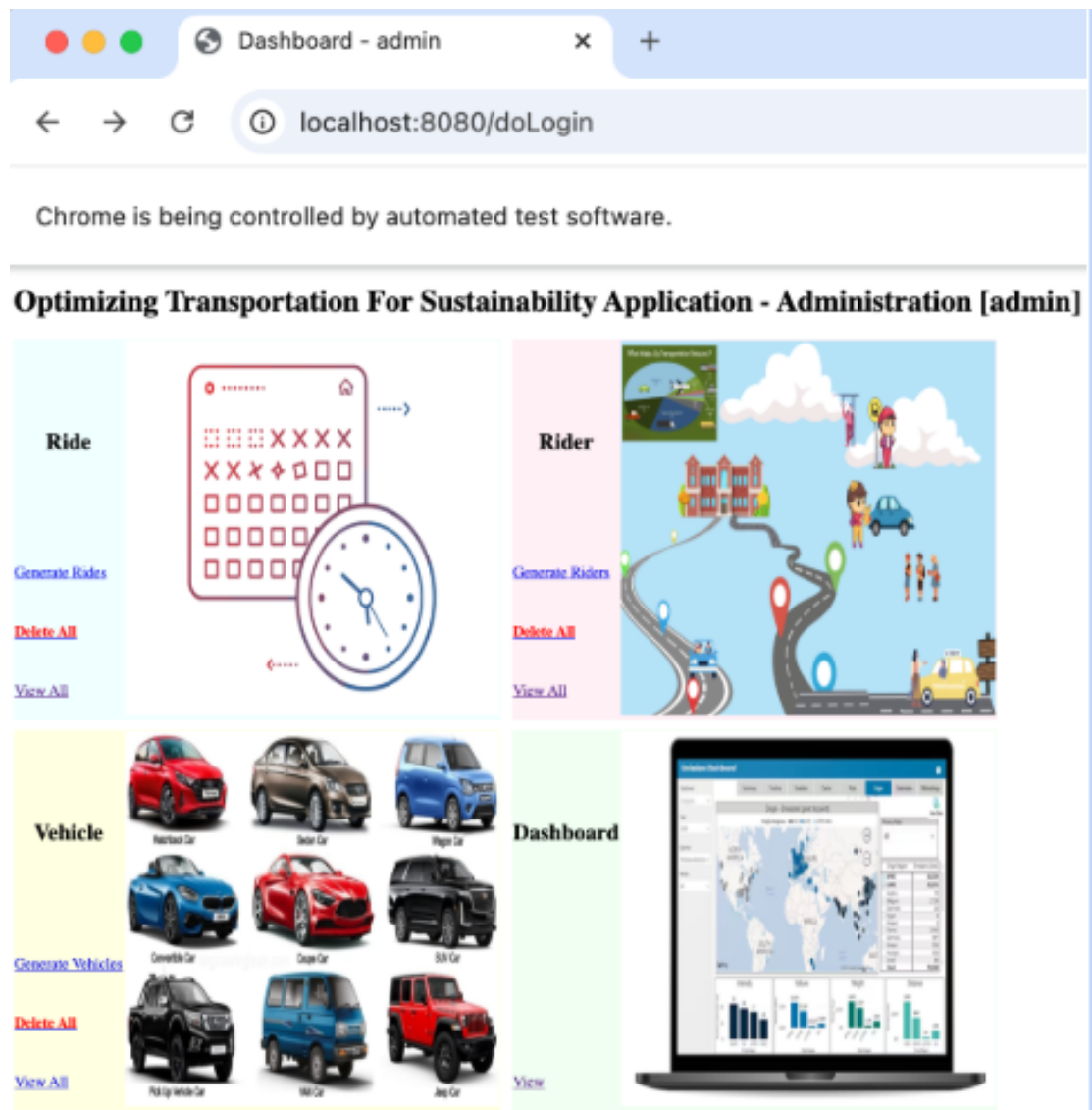
registrationNumber	owner	type	fuelType	co2EmissionInKgsPerKm	capacity
MH 12 SU 1234	test_selenium_user@pictctr.onmicrosoft.com	SUV	Hybrid	0.007	7
OV12 BN 1234	test_selenium_user@pictctr.onmicrosoft.com	Bicycle	None	0.0	2

- **View Riders Selenium Test**

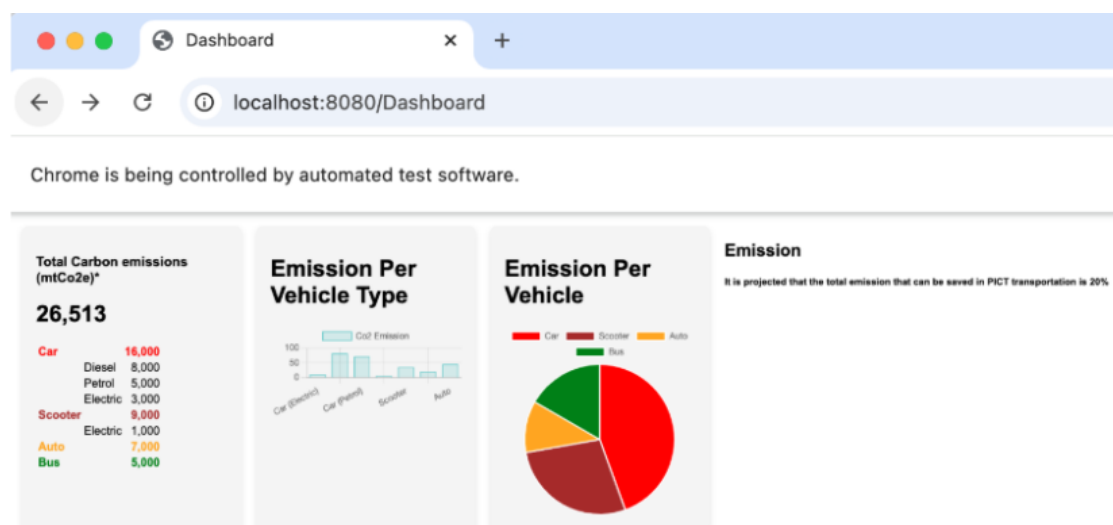


email	name	password	gender	type	branch
test_selenium_user@pictctr.onmicrosoft.com	Selenium Test User	a	Female	Office Staff	All
admin@pictctr.onmicrosoft.com	admin	a	Female	Office Staff	All
E2K21031100@ms.pict.edu	ETC Student 100	E2K21031100	Female	Student	Electronics and Telecommunications
C2K21041100@ms.pict.edu	CS Student	C2K21041100	Female	Student	Computer Science

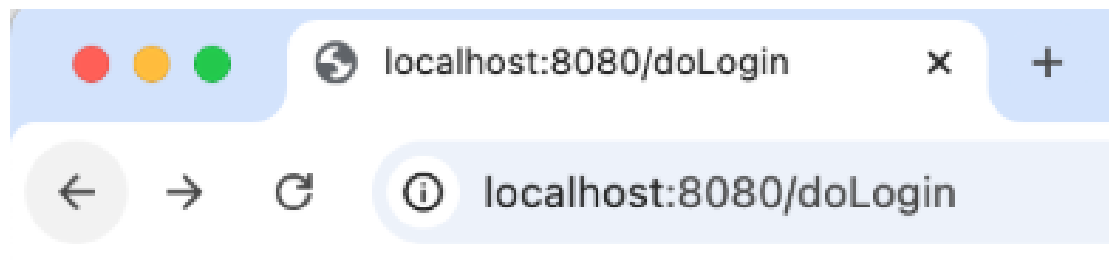
- Admin Login Test with Right Password



- View Dashboard Selenium Test



- Test Admin Login with Wrong Password



Chrome is being controlled by automated test software.

Invalid password for user : admin@pictctr.onmicrosoft.com

- Test View Rides Available for Taking

id	rideOwner	startLocation	endLocation	startTime	vin	distanceInKm	riders
1743562727	staff1@pictctr.onmicrosoft.com	wanawadi	pict	2025-04-02T08:28	OV12 AD 3456	9.5	
1743562797	staff2@pictctr.onmicrosoft.com	hadapsur	pict	2025-04-02T08:29	OV12 AC 4567	14.7	
1743563616	staff1@pictctr.onmicrosoft.com	wanawadi	pict	2025-04-04T08:43	OV12 AD 3456	9.5	staff7@pictctr.onmicrosoft.com E2K21031102@ms.pict.edu
1743571846	E2K21031101@ms.pict.edu	Wanawadi	pict	2025-04-02T11:00	SV12 SE 1234	9.5	
1743578438	E2K21031101@ms.pict.edu	wanawadi	pict	2025-04-04T12:50	SV12 SE 1234	9.5	C2K21041101@ms.pict.edu
1743588821	E2K21031102@ms.pict.edu	Katraj	pict	2025-04-10T15:43	SV12 SP 3456	3.1	E2K21031104@ms.pict.edu
1743598038	C2K21041104@ms.pict.edu	wanawadi	pict	2025-04-16T16:16	SV12 SE 7890	9.5	I2K21051104@ms.pict.edu
1743592110	I2K21051103@ms.pict.edu	Wanawadi	pict	2025-04-16T16:38	SV12 SE 6789	9.5	E2K21031104@ms.pict.edu

- View Vehicles Test

registrationNumber	owner	type	fuelType	co2EmissionInKgsPerKm	capacity
MH 12 SU 1234	test_selenium_user@pictctr.onmicrosoft.com	SUV	Hybrid	0.007	7
OV12 BN 1234	test_selenium_user@pictctr.onmicrosoft.com	Bicycle	None	0.0	2

- JUnit Test Console

```
• vishakhanishra@Vishakhas-MacBook-Air OptimizeTransportationForSustainability % mvn test
[INFO] Scanning for projects...
[INFO]
[INFO] -----< edu.vm.otfs:otfs >-----
[INFO] Building Optimize Transportation For Sustainability 1
[INFO] from pom.xml
[INFO] -----[ war ]-----
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ otfs ---
[INFO] Copying 1 resource from src/main/resources to target/classes
[INFO] Copying 17 resources from src/main/resources to target/classes
[INFO]
[INFO] --- compiler:3.8.1:compile (default-compile) @ otfs ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ otfs ---
[INFO] skip non existing resourceDirectory /Users/vishakhanishra/data/gitrepo/OptimizeTransportationForSustainability/src/test/resources
[INFO]
[INFO] --- compiler:3.8.1:testCompile (default-testCompile) @ otfs ---
[INFO] Changes detected - recompiling the module!
[INFO] Compiling 1 source file to /Users/vishakhanishra/data/gitrepo/OptimizeTransportationForSustainability/target/test-classes
[INFO]
[INFO] --- surefire:3.5.2:test (default-test) @ otfs ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnit4PlatformProvider
[INFO]
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running edu.vm.otfs.SeleniumTest
Apr 11, 2025 9:43:22 AM org.openqa.selenium.devtools.CdpVersionFinder findNearestMatch
WARNING: Unable to find an exact match for CDP version 135, returning the closest version; found: 134; Please update to a Selenium version that supports CDP version 135
Test View Vehicles passed (internally called)
Test View Vehicles passed
Test View Riders passed
Test AdminLoginWithWrongPassword passed
Test Register Vehicle passed
Test View Dashboard passed
Test AdminLoginWithCorrectPassword passed
Test View Riders passed (internally called)
Test View Riders passed (internally called)
Test pre Sign-Up passed
Test View Vehicles passed (internally called)
Test pre Register Vehicle passed (internally called)
[INFO] Tests run: 8, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 30.15 s -- in edu.vm.otfs.SeleniumTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 8, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 38.342 s
[INFO] Finished at: 2025-04-11T09:43:30+05:30
[INFO]
vishakhanishra@Vishakhas-MacBook-Air OptimizeTransportationForSustainability %
```

CHAPTER 8

RESULTS

8.1 OUTCOMES

The successful implementation of the project led to a solution that holistically determines how a ride could be taken, keeping the sustainability aspects of the environment in mind.

8.2 SCREENSHOTS

- Sign-up Page

The screenshot shows a web browser window with the URL `localhost:8080/SignUp?email=demouser%40pictctr.onmicrosoft.com&passwo...`. The page title is "Optimize Transportation Sustainability Application - Sign Up!". The form includes the following fields:

- Full Name:** A text input field with the placeholder "Enter Your Full Name".
- Gender:** A dropdown menu with "Female" selected.
- E-Mail:** A text input field with the value "demouser@pictctr.onmicrosoft.com".
- User Type:** A dropdown menu with "Faculty" selected.
- Password:** A text input field with masked characters.
- Branch:** A dropdown menu with "Computer Science" selected.

A green "Register" button is located at the bottom of the form. To the right of the form is an illustration of a city skyline with a car, a person using a smartphone, and various icons representing connectivity and sustainability.

- Login Page

The screenshot shows a web browser window with the URL `localhost:8080`. The page title is "Optimize Transportation For Sustainability Application - Login!". The form includes the following fields:

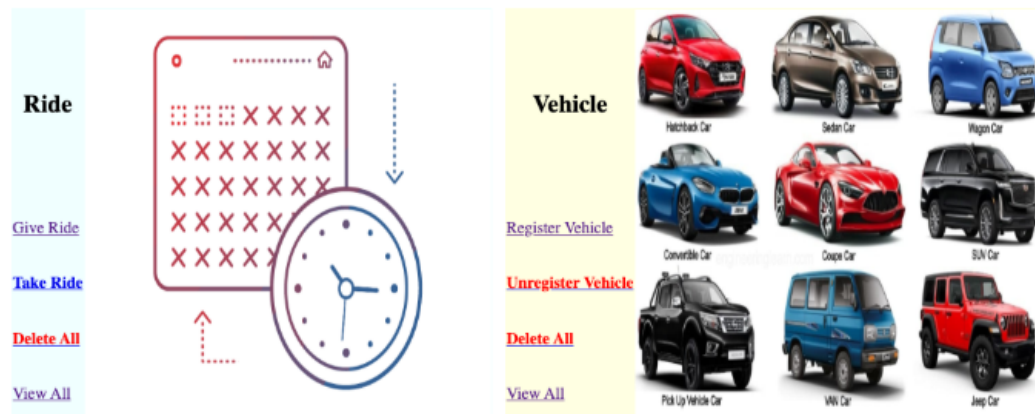
- E-mail id:** A text input field with the value "demouser@pictctr.onmicrosoft.com".
- Password:** A text input field with masked characters.

A green "Login" button is located below the password field. There is a "Remember me" checkbox and a "Forgot password?" link. A blue padlock icon is positioned above the email field. To the right of the form is an illustration of a road winding through a landscape with buildings, trees, and a car, symbolizing sustainable travel.

- Successful Login Landing Page

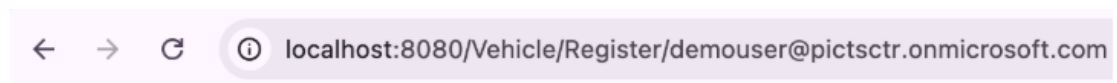


Optimizing Transportation For Sustainability Application - Rider [demo user]



For your effort towards sustainability you have earned 0 🏆 Points

- Register Vehicle



Optimizing Transportation For Sustainability Application - Register Vehicle [demo user]

Vehicle Registration Number:	DV12CE5678
Vehicle Type:	Car
Fuel Type:	Electric
Capacity:	5
CO2 Emission in Kgs per Km:	0.11002
Register	

- Successful Registration Page



- View Registered Vehicles

registrationNumber	owner	type	fuelType	co2EmissionInKgsPerKm	capacity
DV12CE5678	demouser@pictsctr.onmicrosoft.com	Car	Electric	0.11002	5

- Create Ride Page

localhost:8080/Ride/Register/demouser@pictctr.onmicroso...

Optimize Transportation For Sustainability Application - Create Ride [demo user]

Start Location
wanwadi

End Location
pict

Estimated Start Date Time
11/04/2025, 15:17

Set Recurrence on Every
Mon ☐ Tue ☐ Wed ☐ Thu ☐ Fri ☐ Sat ☐ Sun ☐

Estimated End Date dd/mm/yyyy

Vehicle: Car - Electric - DV12CE5678

Register Ride

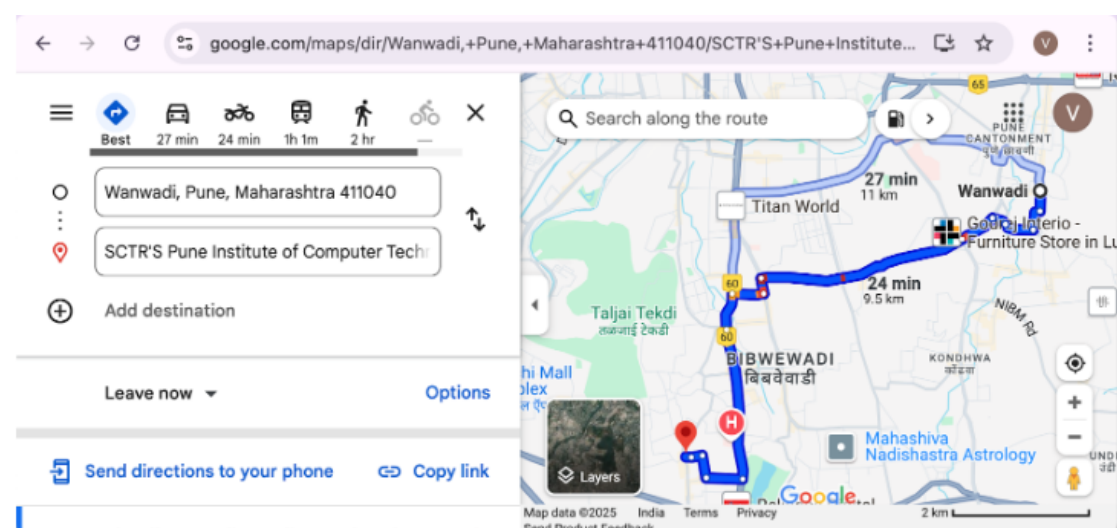
- Successful Ride Creation Page

localhost:8080/doCreateRide

You have created the following 1 ride(s) starting from [wanwadi] to [pict] with google map URL : [map](#)

You have earned 19 🏆 points

- Google Map Page Link on Successful Ride Creation

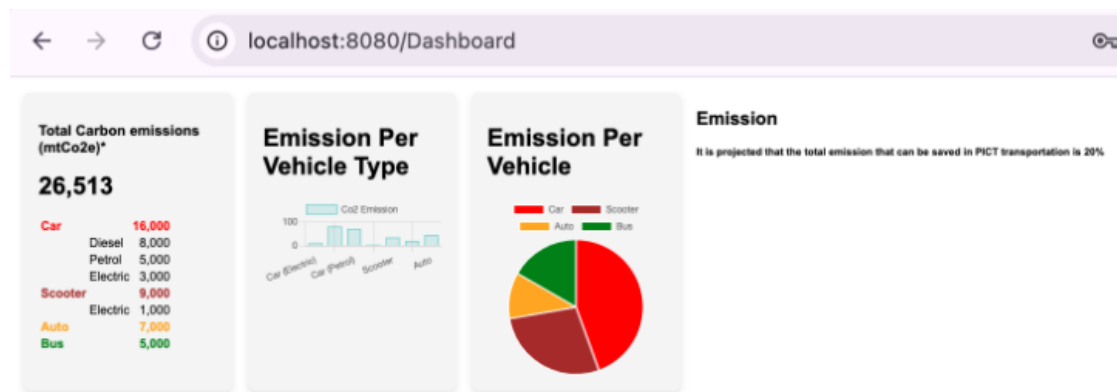


- View Rides Page

localhost:8080/Rides/demouser@pictctr.onmicrosoft.com

id	rideOwner	startLocation	endLocation	startTime	vin	distanceInKm	riders
1284346901	demouser@pictctr.onmicrosoft.com	wanwadi	pict	2025-04-11T15:17	DV12CE5678	9.5	

- Admin Login Dashboard



- Carbon Estimator

Carbon Estimator

Trips per week: 10

Distance per trip (km): 25

Vehicle Efficiency (km/l): 12

Emission Factor (g/km): 4

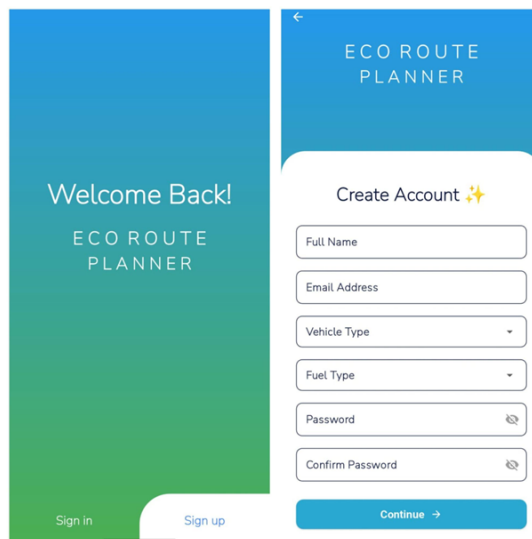
Calculate

Weekly CO₂ Emitted: 1000.00 grams

Weekly Fuel Consumption: 20.83 Liters

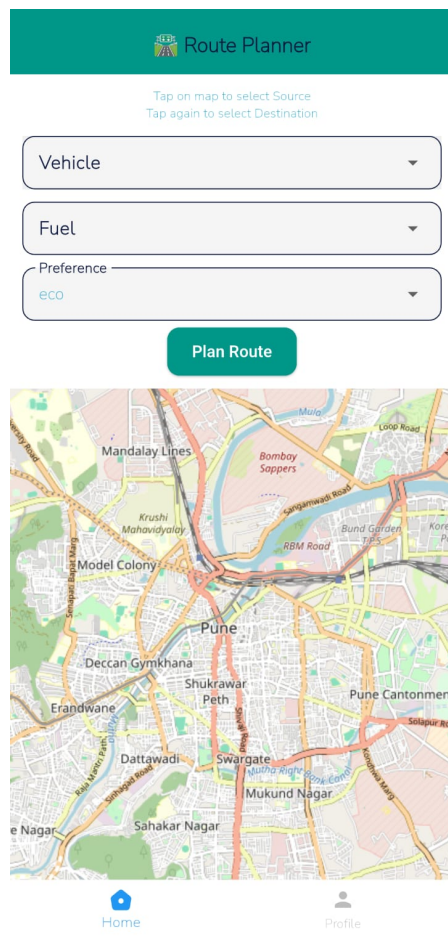
Home Profile

- **Eco-Route Planner Landing Page**



The image shows the landing page of the Eco-Route Planner app. It is split into two main sections. The left section has a green-to-blue gradient background with the text 'Welcome Back!' and 'ECO ROUTE PLANNER' in white. At the bottom of this section are two buttons: 'Sign in' and 'Sign up'. The right section has a white background with a blue header that says 'ECO ROUTE PLANNER'. Below the header is a 'Create Account' button with a star icon. Underneath are several input fields: 'Full Name', 'Email Address', 'Vehicle Type' (a dropdown menu), 'Fuel Type' (a dropdown menu), 'Password', and 'Confirm Password' (both with eye icons to toggle visibility). At the bottom of the right section is a blue 'Continue' button with a right-pointing arrow.

- **Route Planner**



The image shows the 'Route Planner' interface of the app. At the top is a green header with the app's logo and the text 'Route Planner'. Below the header are two lines of instructional text: 'Tap on map to select Source' and 'Tap again to select Destination'. Underneath are three dropdown menus: 'Vehicle', 'Fuel', and 'Preference' (which is currently set to 'eco'). Below these menus is a green 'Plan Route' button. The main part of the screen is a map of Pune, India, showing various roads and landmarks. At the bottom of the screen is a navigation bar with two icons: a house icon labeled 'Home' and a person icon labeled 'Profile'.

CHAPTER 9
CONCLUSIONS

9.1 CONCLUSIONS

The project "Optimizing Transportation For Sustainability" presents a complex problem that intertwines computational complexity and sustainability goals. The feasibility assessment suggests that while the core problem of finding the shortest path is a P problem, the introduction of multiple sustainability constraints likely elevates it to NP-complete or NP-hard status. The use of advanced algorithms, mathematical modeling, and real-time data integration can provide actionable insights for urban planners and policymakers aiming to reduce carbon emissions and enhance transportation efficiency.

The ride-sharing application ultimately provides the following benefits:

- **Enhanced Efficiency:** By optimizing transportation routes, the system reduces travel times and fuel consumption, leading to increased operational efficiency.
- **Environmental Benefits:** The integration of sustainability metrics helps in lowering carbon emissions, contributing to a greener environment.
- **User-Friendly Interface:** The web-based application is designed to be intuitive, making it accessible for a wide range of users, including tech-savvy individuals and urban planners.
- **Data-Driven Insights:** Provides actionable insights and detailed reports for policymakers and urban planners, enabling informed decision-making regarding transportation infrastructure.
- **Real-Time Updates:** The system utilizes real-time data from various sources, allowing users to make informed decisions based on current traffic conditions and emissions data.

9.2 FUTURE WORK

The future scope of work involves the following:

- Implementing Sustainability score of a route in Google Map.
- Evaluating when to remove hazardous vehicles (end-of-life) and routes for maintenance.
- Creating zones that spread our traffic to improve overall AQI (Air Quality Index).
- Allowing end-users to automatically pick rides based on the nearest location from their current coordinates.
- Drawing deep insights from Google Map API for real-time ride tuning.

9.3 APPLICATIONS

- **Urban Transportation Planning:** The system can be used by city planners to analyze transportation patterns and develop strategies for more sustainable urban mobility.
- **Ridesharing Services:** Ride-sharing companies can integrate this system to optimize routes for drivers and riders, enhancing efficiency and reducing costs.
- **Environmental Research:** Researchers studying transportation-related emissions can utilize the system to gather data and analyze trends in urban mobility.
- **Policy Formulation:** Policymakers can use insights generated by the system to create informed policies that promote sustainable transportation practices.
- **Logistics and Delivery Services:** Companies in logistics can leverage route optimization to reduce delivery times and costs while minimizing their carbon footprint.

CHAPTER 10

APPENDIX

10.1 APPENDIX A: Problem Statement Feasibility Assessment Using Satisfiability Analysis and NP-Hard, NP-Complete or P Type Using Modern Algebra and Relevant Mathematical Models

6.1.1 Introduction

This report assesses the feasibility of the problem statement concerning the optimization of transportation for sustainability, utilizing modern algebra and relevant mathematical models. The project aims to integrate advanced algorithms, particularly the A* algorithm, with sustainability metrics to optimize transportation routes. Given the complexities involved, this analysis evaluates the problem's classification in terms of computational complexity (P, NP-Complete, NP-Hard) and its satisfiability through mathematical modeling.

6.1.2 Problem Definition

The primary objective is to find the shortest path between a start node s and a goal node t in a transportation network while accounting for sustainability factors such as travel time, emissions, and fuel consumption. This can be represented mathematically as:

Graph Representation:

- $G = (V, E)$ where V is the set of nodes (intersections, points of interest) and E is the set of edges (roads connecting the nodes).
- Each edge (i, j) has associated costs T_{ij} (travel time), E_{ij} (emissions), and F_{ij} (fuel consumption).

Cost Function:

$$C_{ij} = \alpha \cdot T_{ij} + \beta \cdot E_{ij} + \gamma \cdot F_{ij}$$

where:

- T_{ij} is the cost of travel time,
- E_{ij} is the cost of emission,
- F_{ij} is the cost of fuel consumption,
- α , β , and γ are weighting factors for travel time, emissions, and fuel consumption, respectively.

6.1.3 Satisfiability Analysis

The problem can be analyzed using satisfiability concepts from propositional logic, specifically focusing on the existence of a feasible path that meets sustainability criteria. This involves:

- Defining constraints that any feasible solution must satisfy, such as:
 - Total emissions must not exceed a specified threshold.
 - Travel time should be minimized while ensuring that fuel consumption remains within acceptable limits.

6.1.4 Computational Complexity Assessment

To classify the problem, we consider:

- **P (Polynomial Time):** Problems solvable in polynomial time. Finding a single shortest path in a weighted graph is an example of a P problem, typically solvable using Dijkstra's or A* algorithms.
- **NP (Nondeterministic Polynomial Time):** Problems for which a solution can be verified in polynomial time. The proposed optimization problem could involve multiple constraints (e.g., minimizing emissions while maximizing efficiency) that may require verification of multiple paths.
- **NP-Complete:** Problems that are both in NP and NP-hard. The challenge of optimizing paths based on multiple criteria (travel time, emissions, fuel consumption) could elevate the problem to NP-complete status, particularly if it involves subsets or combinations of paths.
- **NP-Hard:** A classification for problems that are at least as hard as the hardest problems in NP. If the problem were generalized to involve additional constraints, such as varying vehicle characteristics and dynamic traffic conditions, it could be NP-hard.

6.1.5 Mathematical Models

Using graph theory and the A* algorithm, we can formulate the following mathematical components:

- **Cost Function Optimization:** The goal is to minimize the combined cost function:

$$C = \sum_{i,j \in P} C_{ij}$$

where P represents the chosen path from s to t .

- **Dynamic Weighting:** The weights α , β , and γ can be adjusted dynamically based on real-time data, reflecting current traffic conditions and environmental factors.
- **Constraint Satisfaction:** Incorporating constraints:

$$E \leq E_{max}, \quad T \leq T_{max}, \quad F \leq F_{max}$$

where E , T , and F represent emissions, travel time, and fuel consumption, respectively.

10.2 APPENDIX B: DETAILS OF PAPER PUBLICATIONS

10.3 APPENDIX C: PLAGIARISM REPORT

11 REFERENCES

1. Banister, D. (2008). The sustainable mobility paradigm. *Transport Policy*, 15(2), 73-80.
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